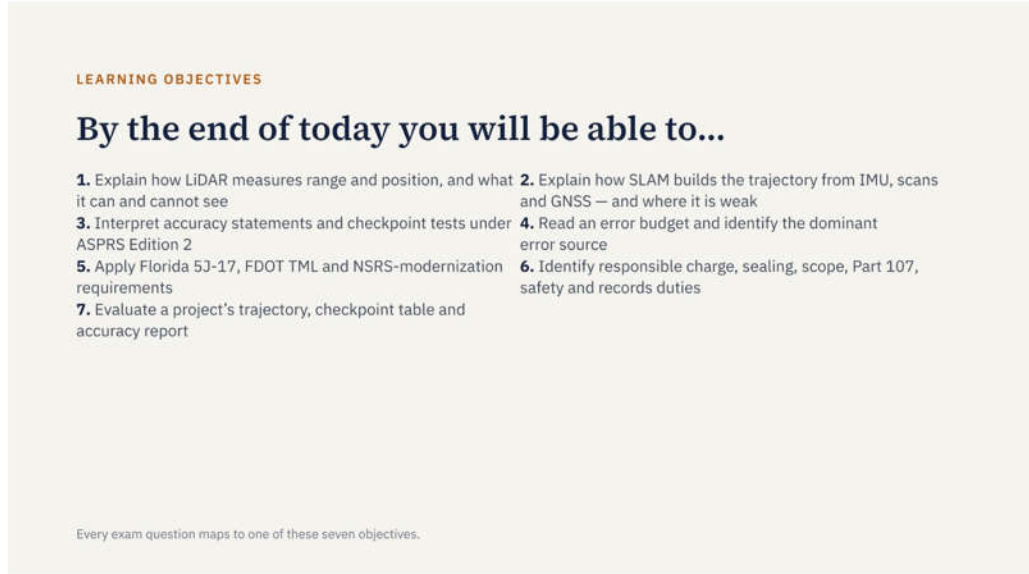


Welcome everyone. This seminar is about mobile LiDAR from a surveyor's point of view: how the measurement works, how SLAM builds the trajectory — the core of the day — where the errors come from, how to test and report accuracy under ASPRS Edition 2, and what Florida's Standards of Practice expect when you sign and seal LiDAR deliverables. You are responsible for the graphs and the concepts, not the equations. The book is your reference after today; the slides carry the day. The cover image is a simulated street point cloud colored by height.

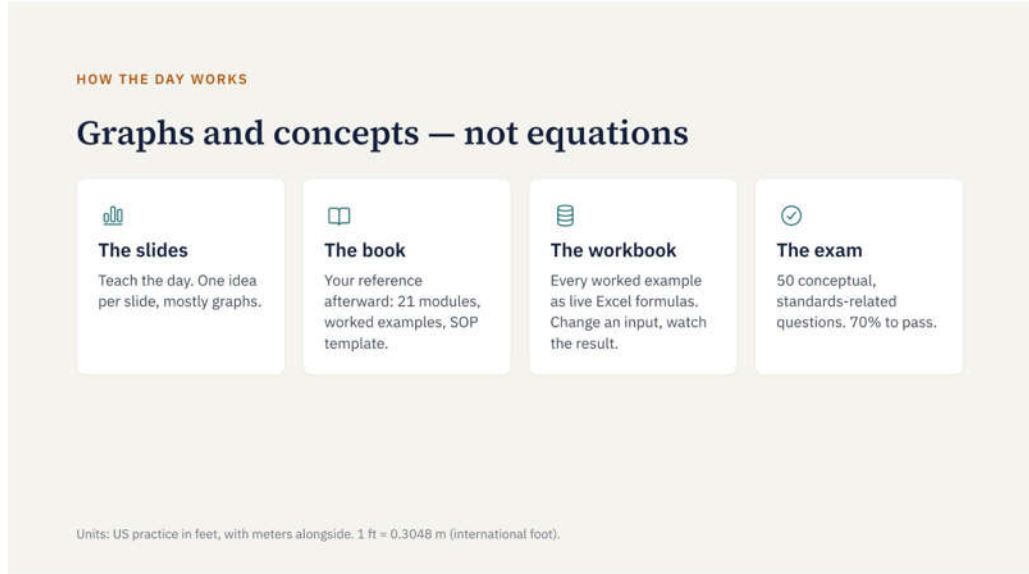


Read the objectives aloud quickly. Point out that the exam questions are tagged to these seven objectives in the instructor packet, so nothing on the exam is outside this list. Emphasize objective 2: SLAM is the core of the day and gets the longest block. Objective 4 — reading an error budget — is the most useful skill for judging a LiDAR proposal.

Today's agenda

Time	Block	Modules
0:00–0:10	Welcome, objectives, how the day works	—
0:10–0:40	1 · How LiDAR measures; what it can see through	1, 3, 4, 5
0:40–1:15	2 · Pose, IMU, GNSS, timing, calibration	6, 7, 8, 9
1:15–1:30	Break	
1:30–2:40	3 · SLAM: building the trajectory (core block)	10, 11, 12
2:40–3:10	Lunch	
3:10–3:55	4 · ASPRS accuracy and error budgets	13, 14
3:55–4:25	5 · Case studies: OS0, GX5/GQ7, RIEGL	15, 16, 17
4:25–4:40	6 · Processing and QA/QC	18
4:40–5:20	7 · Florida standards, datums, professional practice	19, 20
5:20–5:40	8 · Capstone dataset walk-through	21
5:40–6:10	Final exam and evaluation	packet

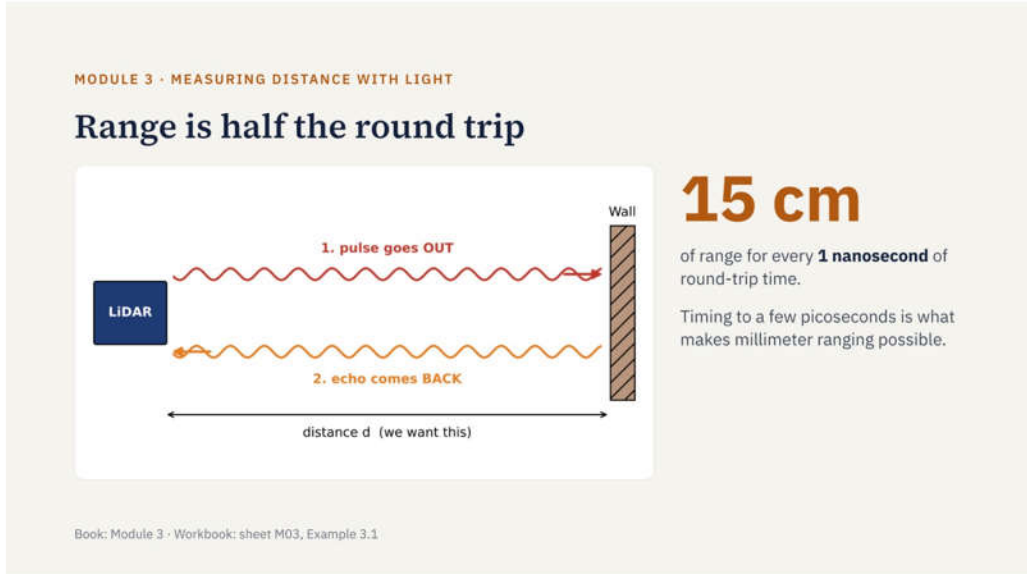
Instruction time is 4 hours 55 minutes; breaks 45 minutes; exam 30 minutes. Confirm the contact hours the Board will credit before advertising — adjust block lengths if you need a full 5 or 6 instructional hours. SLAM is the core of the day: Block 3 runs 70 minutes, the longest block, and every other block references it.



Set expectations: nobody will be asked to compute anything on the exam. Engineers who want the math will find every equation explained in the book and every worked example rebuilt in the Excel workbook. Mention the international foot now — it comes back in Block 7.



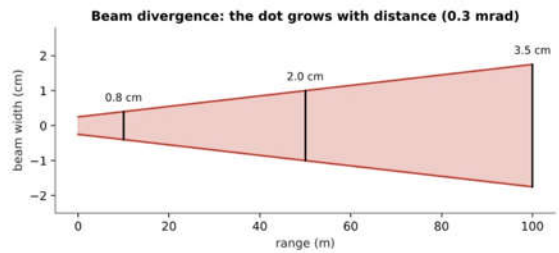
30 minutes. Goal of the block: participants leave able to explain why a LiDAR point is not a pin-point but a blurred spot, why dark surfaces shorten range, and exactly what the laser can and cannot see through.



Light covers about 30 cm per nanosecond; because the pulse goes out and back, each nanosecond is 15 cm of range. A 400 ns echo is about 60 m. Air slows light by about 300 parts per million, which software corrects from temperature and pressure — that comes back in the RIEGL case study, where 300 ppm at 500 ft is 4.6 cm. Mention multiple pulses in the air: faster firing shortens the unambiguous range.

MODULE 4 · THE BEAM AND THE ECHO

A point is really a spot



The beam spreads. At 100 m (330 ft) a 0.3 mrad beam is a **3.5 cm (0.11 ft)** spot.

An edge — a curb lip, a pole — can only be located to about **half the spot size**.

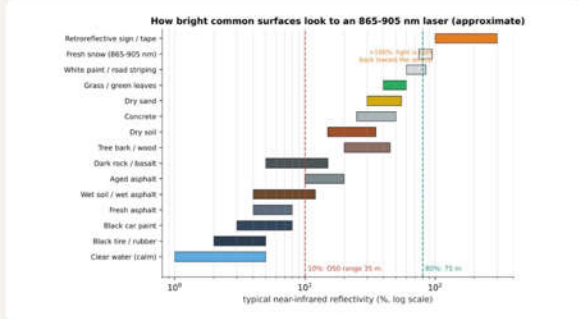
Fine detail is best captured close to the sensor.

Book: Module 4, Example 4.1 · $D = d_0 + \rho\cdot\gamma$

Footprint equals exit diameter plus range times divergence. 5 mm exit and 0.3 mrad gives 2 cm at 50 m and 3.5 cm at 100 m. Ask the room: if you need curb lines to plus or minus 1 cm, how close does the scanner need to be? Answer: close — and that is why mobile mapping beats airborne for hardscape detail.

MODULE 4 · REFLECTIVITY

Dark surfaces send back less light



Fresh asphalt, black paint and tires reflect **under 10%**.

White striping and retroreflective signs reflect far more — sometimes too much.

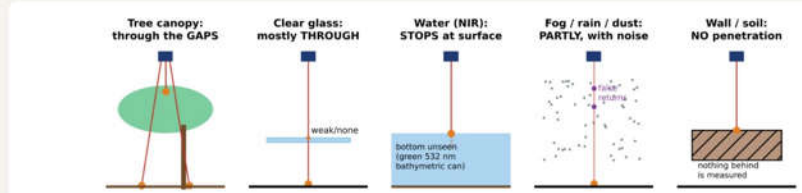
Values are typical; wetness and angle change them.

Book: Module 4, Section 4.3

Walk down the chart from bright to dark. Point out the two dashed lines: the Ouster OS0 is specified to 35 m on a 10% target and 75 m on an 80% target. New asphalt sits around 5%. Wet surfaces get darker. Retroreflective signs can saturate the receiver and create range errors or blooming.

MODULE 4 · PENETRATION

LiDAR sees through gaps, not through solids



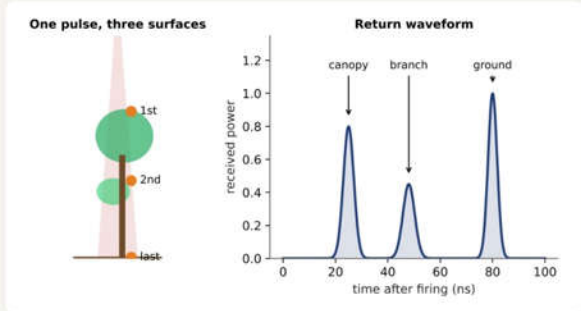
Canopy	Glass	Water (NIR)	Fog / dust	Wall / soil
Through the gaps	Mostly through	Stops at surface	Partly, noisy	No

Book: Module 4, Section 4.4 · State these limits in the Survey and Map Report

This is the slide clients most often misunderstand. LiDAR gets ground under trees only through gaps between leaves — more returns, more chances. It does not see under water with a near-infrared laser (green bathymetric lasers can, in clear water). It does not see underground utilities. Fog and dust create false returns that must be filtered. Every one of these belongs in the limitations section of your report.

MODULE 4 · MULTIPLE RETURNS

One pulse, several echoes



Part of the spot hits leaves, part reaches the ground: the receiver records **first, intermediate and last** returns.

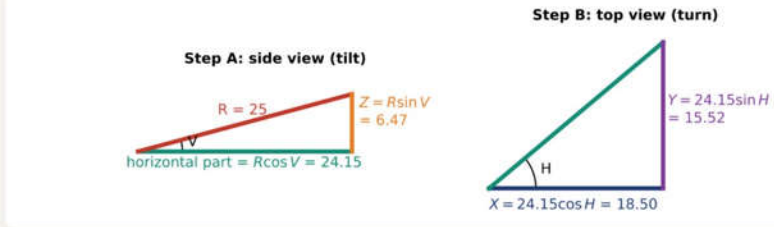
The last return is the best candidate for ground — but not always ground.

Book: Module 4 · LAS 1.4 stores return number and number of returns for every point

Connect to vegetated vertical accuracy (VVA) in Block 4: under vegetation the last return can land on low shrubs, so VVA is tested separately from non-vegetated accuracy. The OS0 records up to 2 returns; airborne survey scanners record many more.

MODULE 5 · FROM ANGLES TO X, Y, Z

A scanner is a very fast total station

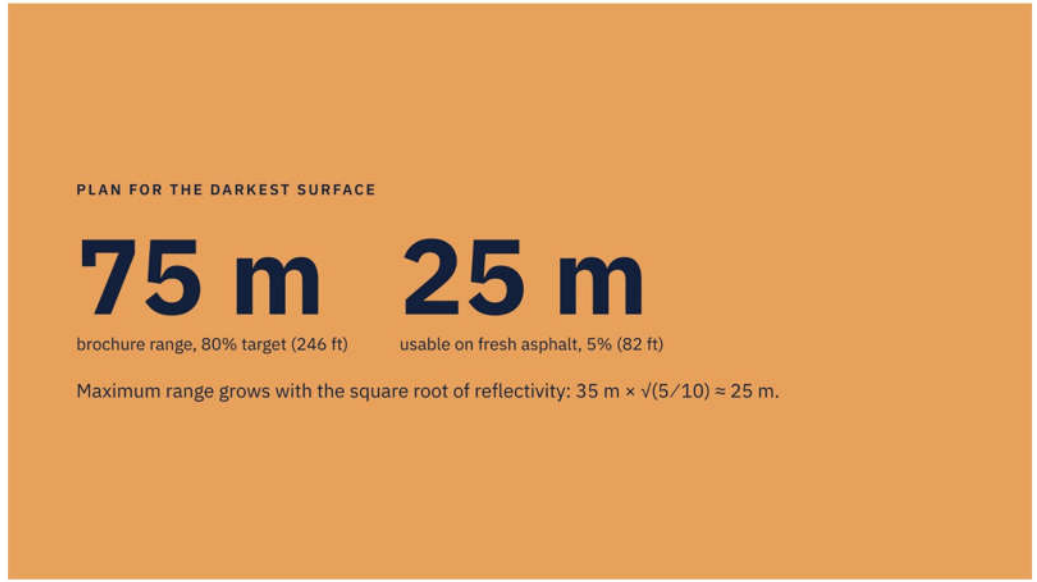


Each point = **range + two angles**, exactly like a total-station shot — hundreds of thousands per second.

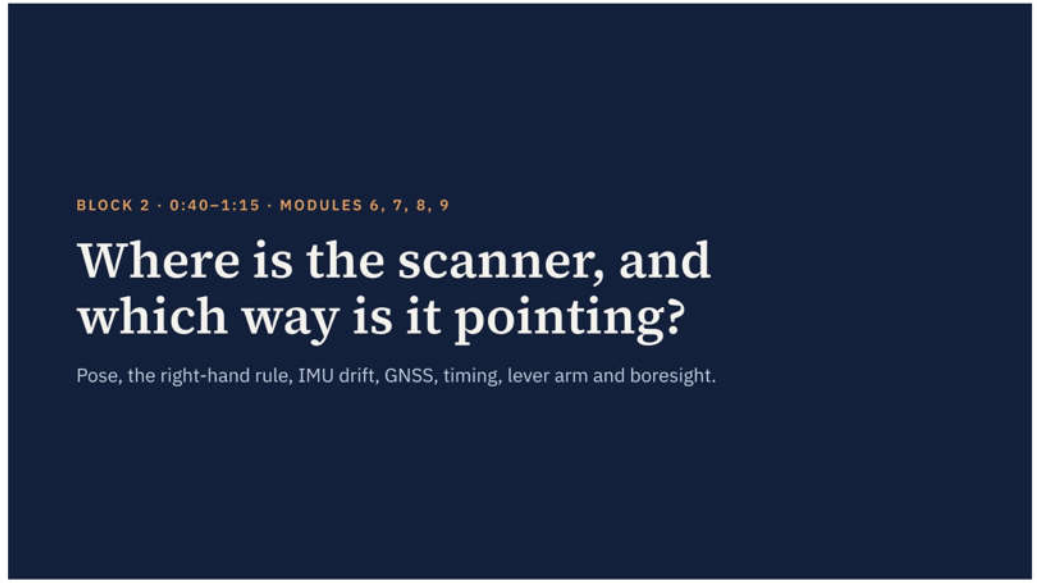
Spacing grows with range: **along-track = speed ÷ rotation rate**; across-track = range × angular step.

Book: Module 5, Examples 5.1–5.2 · Workbook sheet M05

Surveyors already know this geometry — it is a total station shot in the sensor's own frame. Example: 16 channels at 10 Hz driving 15 m/s gives 1.5 m (4.9 ft) between scan lines: fine for pavement, too coarse for curbs. Slow down, tilt the head, or add channels. In the workbook, change speed on sheet M05 and watch spacing change.



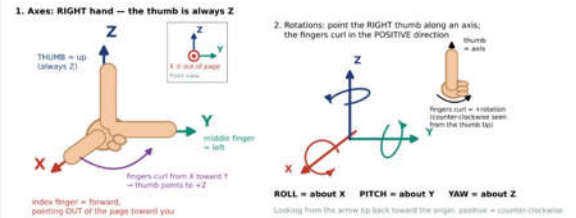
This is the takeaway of Block 1. A sensor sold as a 75 m scanner may deliver road coverage only to about 25 m on new asphalt. Size the swath, the number of passes and the drive line to the darkest surface you must map. Ask: what is the darkest surface on your next project? Transition to Block 2 — knowing where each point is relative to the scanner is only half the problem; next we need to know where the scanner is.



35 minutes. A mobile scanner is a total station that never stops moving. Every point needs the pose — position and orientation — at the instant it was measured. This block covers where that pose comes from and how small angle and timing errors become centimeters on the ground.

MODULE 6 · FRAMES

The right-hand rule: the thumb is always Z



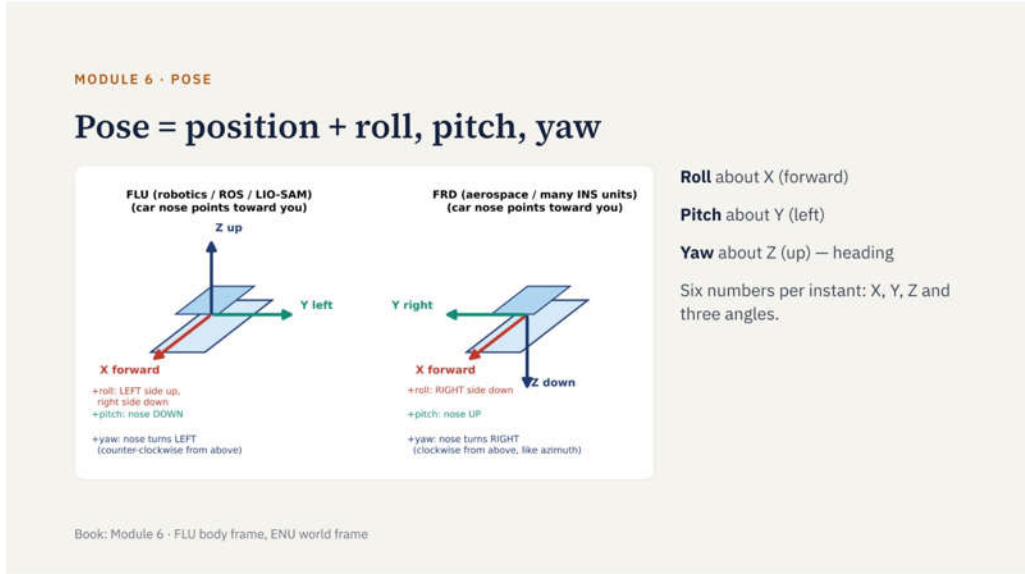
Index = X, middle = Y, thumb = Z.

Point the thumb along an axis and curl your fingers: that is the positive rotation.

"X forward, Y right, Z up" fails the test — it is left-handed.

Book: Module 6, Example 6.1

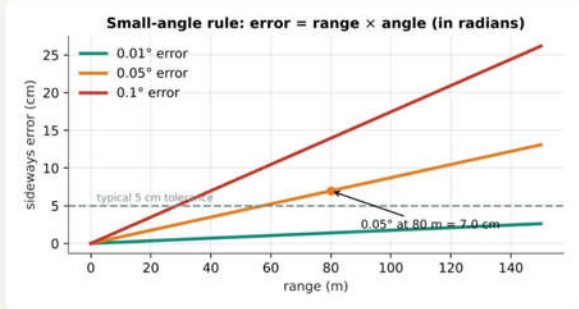
Have everyone hold up their right hand. Index forward, middle finger to the right: where does the thumb point? Down. So a datasheet that says X forward, Y right, Z up is either wrong or means Z down (the FRD convention used in aviation). Resolve it with the manufacturer before computing any lever arm — a sign error mirrors or tilts the whole cloud.



FLU means forward, left, up — the convention used in the book and in ROS-based systems like LIO-SAM. The world frame is east, north, up. The trajectory is simply this six-number pose, recorded hundreds of times per second.

MODULE 2 · THE SMALL-ANGLE RULE

Small angles become big offsets at range



Offset = range × angle (in radians).

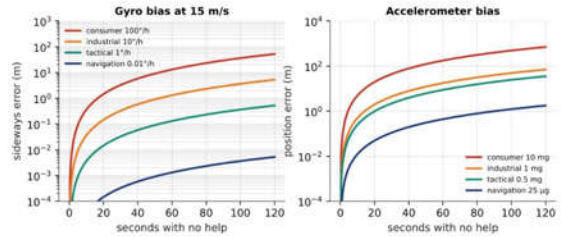
Error	At 100 m (330 ft)
20″	1.0 cm (0.03 ft)
0.05°	8.7 cm (0.29 ft)
0.1°	17.5 cm (0.57 ft)

Book: Modules 2 and 6 · "20 seconds at 100 m is a centimeter"

Every surveyor knows 20 seconds at 100 m is about a centimeter. The same rule governs heading, roll, pitch and boresight errors in LiDAR. A tenth of a degree — which sounds tiny for an IMU — is 17.5 cm at 100 m, bigger than most engineering tolerances. Keep this rule in mind; every error budget later today uses it.

MODULE 7 · THE IMU

An IMU alone drifts — fast



A **tactical** IMU coasting 60 s at 15 m/s:

13 cm sideways from the gyro…

…but **8.8 m (29 ft)** along-track from a 0.5 mg accelerometer bias.

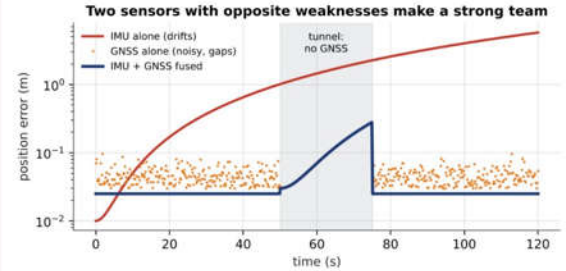
Error grows with t^2 and t^3.

Book: Module 7, Example 7.2 · Workbook sheet M07

The IMU is a superb short-term motion sensor and a hopeless long-term navigator. Accelerometer bias integrates twice, so position error grows with the square of time; gyro bias tilts the platform and leaks gravity, so it grows even faster. A 0.1 degree tilt error leaks 1.7 mg of gravity — worse than the accelerometer bias itself. No survey system coasts on its IMU for long.

MODULES 7–8 · GNSS + IMU

Opposite weaknesses make a strong team



IMU: smooth and fast, but drifts.

GNSS: never drifts, but noisy and blocked by tunnels, trees and buildings.

Fused: the filter takes the best of each — until the outage runs long.

Book: Module 8 · In Florida, RTK corrections are available free from FPRN

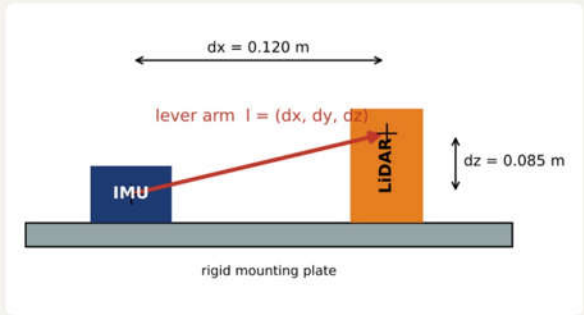
Point at the gray tunnel band: during the outage, the fused solution grows like the IMU; when GNSS returns, it snaps back. The length of the outage and the quality of the IMU decide how far it drifts. In Block 5 we will put numbers on this for three real IMUs.



Every sensor must share one clock. GNSS provides a pulse-per-second and time message; PTP does the same over Ethernet. Without sync, a free-running clock drifting 20 ppm is 12 ms off after 10 minutes — 18 cm at 15 m/s (34 mph). Within a single sweep the vehicle moves 1.5 m, which is why every point is de-skewed using the IMU before scan matching. The diagnostic is the turn signature: if straights agree and corners smear, suspect timing before calibration.

MODULE 9 · CALIBRATION

Lever arm and boresight



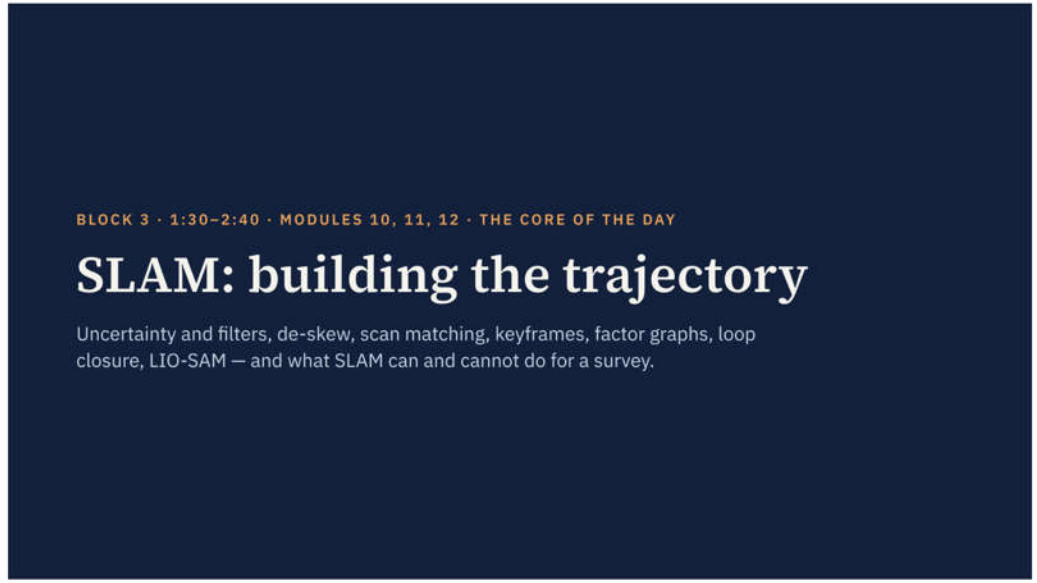
Lever arm = offset between IMU and scanner. A 7 mm error is 7 mm everywhere.

Boresight = angular misalignment. 0.05° is 7 cm at 80 m.

Boresight dominates far away; lever arm close in.

Book: Module 9 · Calibrate after every remount; document it in the measurement records

The lever arm is measured with a total station or from CAD; the boresight is solved from a calibration drive or flight — opposite-direction passes over planar features. Recalibrate after any remount or hard knock and keep the calibration file with the project records — Florida's measurement-records rule, 5J-17.050(8), covers it.



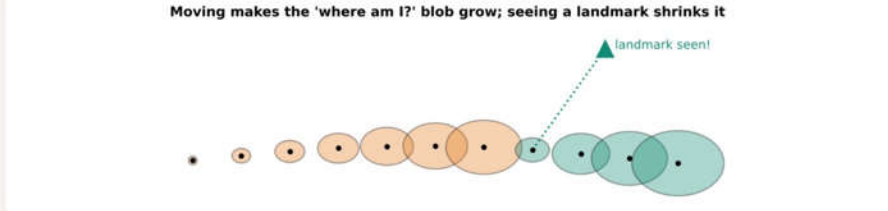
70 minutes — the longest block and the heart of the seminar. Every mobile LiDAR trajectory is built this way. Teach it through survey analogies: the Kalman filter is a weighted mean, scan matching is a best-fit transformation, the factor graph is a least-squares adjustment, and loop closure is closing a traverse. Participants are responsible for the graphs and concepts; the equations and every worked example are in the book and the workbook (sheets M10–M12). Suggested pacing: 20 min filters and uncertainty, 20 min scan matching, 20 min factor graphs and loop closure, 10 min LIO-SAM results, QA and the summary.



This is the core of the day. Every mobile LiDAR system — backpack, vehicle, handheld, many drones — relies on SLAM or SLAM-like scan matching to hold the trajectory where GNSS is weak. Surveyor's framing: it is a resection and a traverse being computed continuously, with the walls and poles as the targets. Over the next hour we follow it piece by piece: uncertainty, filters, scan matching, factor graphs, loop closure, and what it all means for accuracy.

MODULE 10 · UNCERTAINTY

Uncertainty grows, then shrinks



Every step of dead reckoning adds error: the “where am I?” ellipse **grows**.

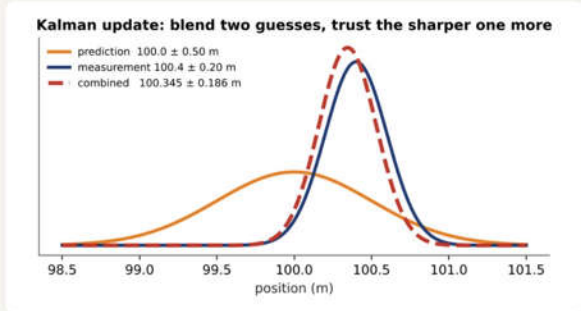
Recognizing a landmark — a pole, a wall corner, a control target — **shrinks** it.

Book: Module 10 · The same idea as a traverse that checks into control

All of SLAM is this cycle: predict (the blob grows) and correct (the blob shrinks). The quality of the trajectory depends on how often good corrections arrive. In a featureless field or a long blank corridor they don’t, and the blob keeps growing — that is where you place control.

MODULE 10 · THE KALMAN FILTER

Predict, then correct



Guess	Value
Prediction	100.0 ± 0.50 m
Measurement	100.4 ± 0.20 m
Combined	100.345 ± 0.186 m

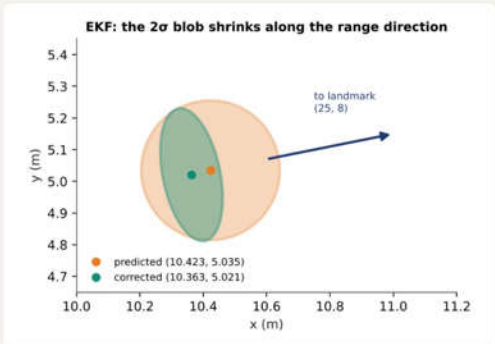
A **weighted mean**: the combined answer beats both inputs. Surveyors know it as least squares.

Book: Module 10, Example 10.2 · Workbook sheet M10

Gain K = 0.25 / (0.25 + 0.04) = 0.86: the measurement is sharper, so it gets 86% of the say. The result, ±0.186 m, is better than either input — the payoff of combining independent information. An EKF is the same thing in several dimensions, with the motion model linearized at each step.

MODULE 10 · EKF WITH A LANDMARK

One range shot tightens one direction



Range and bearing to a pole at (25, 8):

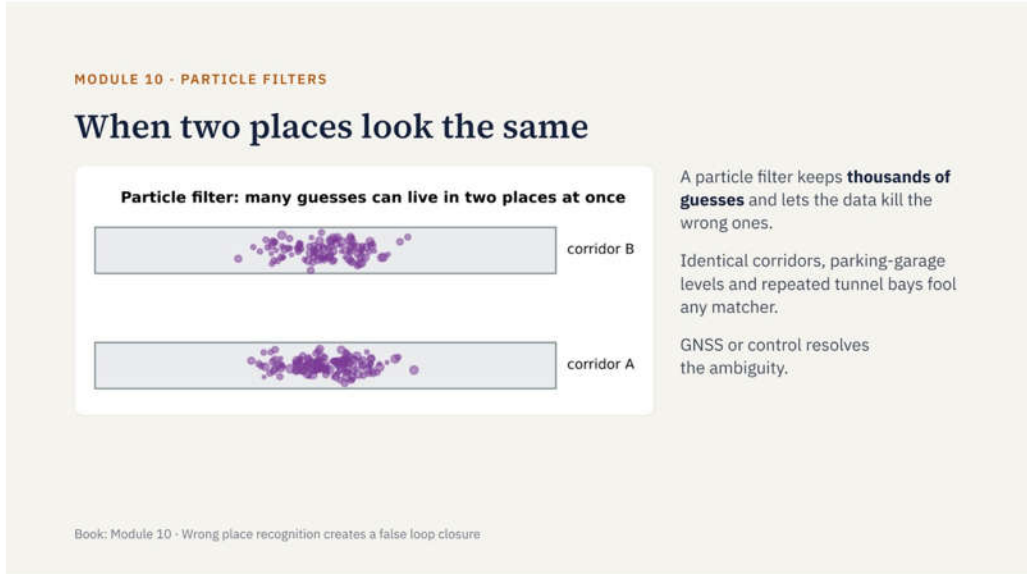
x: **11.0 → 4.9 cm** (along the line of sight)

y: **11.0 → 10.5 cm** (across it — barely helped)

Geometry matters: like a single distance shot, a landmark only constrains the direction it points.

Book: Module 10, Examples 10.3–10.4 · Workbook sheet M10 (full EKF in formulas)

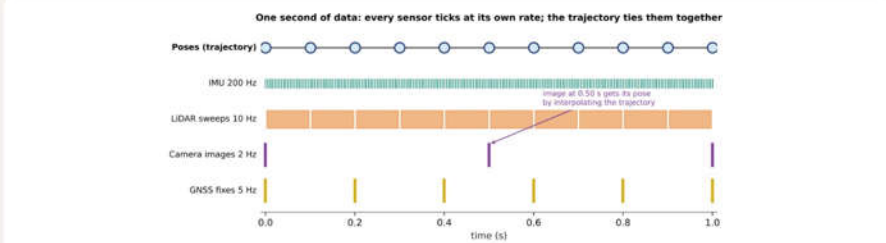
The pole is almost straight ahead, so the range measurement pins forward-back position and the bearing pins heading, but sideways position barely improves. That is strength of figure — the same reason surveyors want intersecting rays at good angles. SLAM accuracy depends on landmark geometry around the vehicle, not just on how many points there are.



Survey relevance: repeated geometry is the main cause of a false loop closure, where the software decides it is back at a place it has never been. The map then folds onto itself. Watch for it in parking garages, identical bridge spans and long uniform buildings, and check the trajectory report for sudden jumps.

MODULE 12 · ONE SECOND OF DATA

Every sensor ticks at its own rate



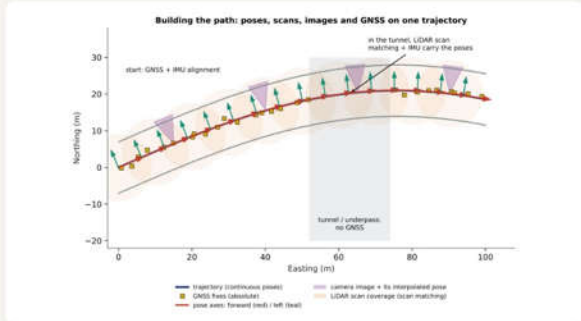
The trajectory is the common thread: every laser point and every photo gets its pose by **interpolating the trajectory at its own timestamp**.

Book: Module 12, Example 12.2 — a photo at 10.04 s gets the pose 40% of the way between 10.00 s and 10.10 s

IMU 200 Hz, LiDAR 10 Hz sweeps, camera 2 Hz, GNSS 5 Hz. None of them line up. The trajectory is estimated as a smooth path, and each measurement is placed on it by its time stamp. This is why timing (Block 2) matters so much — a wrong time stamp reads the wrong pose.

MODULE 12 · THE PATH

Poses, scans, images and GNSS on one path



GNSS anchors the path to the datum.

IMU fills in between fixes.

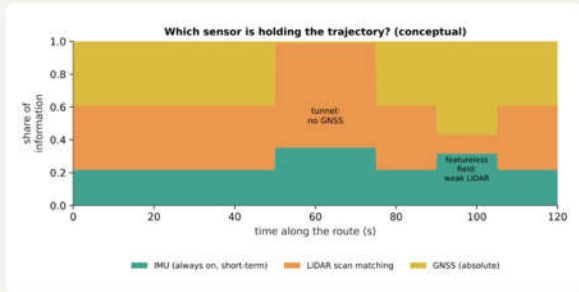
Scan matching carries the poses through the tunnel.

Book: Module 12 · Figure "Building the path"

Read the figure left to right: GNSS plus IMU alignment at the start; fixes along the open road; then the gray tunnel where GNSS disappears and LiDAR scan matching plus the IMU keep the poses. Ask: what would you add in the tunnel to protect accuracy? Answer: control targets — and that is exactly what the capstone does.

MODULE 12 · WHO HOLDS THE TRAJECTORY

Three sensors take turns



Tunnel: GNSS is gone; LiDAR and IMU carry it.

Featureless field: LiDAR is weak; GNSS and IMU carry it.

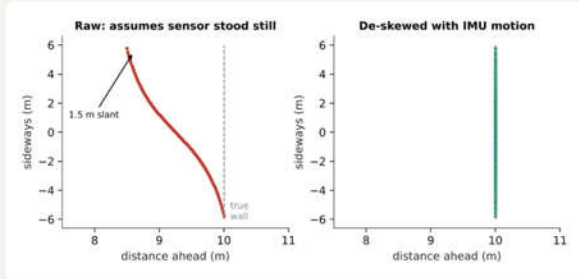
Trouble comes when **two are weak at once**.

Book: Module 12 · Conceptual figure

This is the planning slide for SLAM projects. Walk the route in your head: where is GNSS blocked, and where is there nothing for the LiDAR to grab? A tunnel with smooth walls is both — that is where control targets and extra checkpoints go. The IMU is always on, but only for seconds.

MODULE 8 · DE-SKEW

Step one of every scan: de-skew



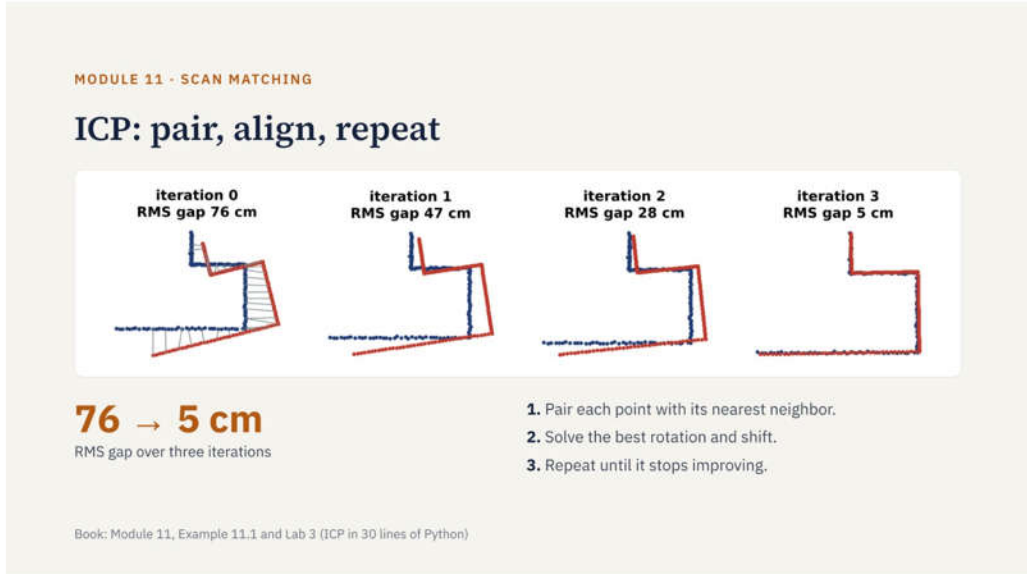
A 10 Hz sweep at 15 m/s: the sensor moves **1.5 m** while it spins.

The IMU says how it moved; each point is shifted to one instant.

Matching skewed scans gives a wrong pose.

Book: Module 8, Example 8.2 · LIO-SAM de-skews with IMU-predicted motion before matching

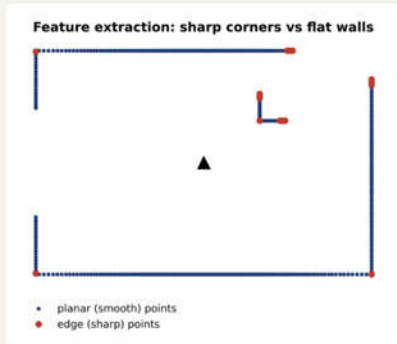
Without de-skewing, a straight wall is drawn as a slant and the scan matcher tries to fit a distorted shape — the pose comes out wrong. This is the first place the IMU and the LiDAR work together inside SLAM, and why timing sync from Block 2 is a SLAM requirement, not a nicety.



ICP — iterative closest point — is a best-fit transformation, like fitting one set of control coordinates onto another, except the point pairs are only guesses that improve each round. A good starting guess from the IMU is essential: start too far off and ICP converges to the wrong answer. Point-to-plane matching converges faster on walls and pavement than point-to-point.

MODULE 11 · LOAM FEATURES

Match only edges and planes



Each point gets a **smoothness** score from its neighbors' ranges.

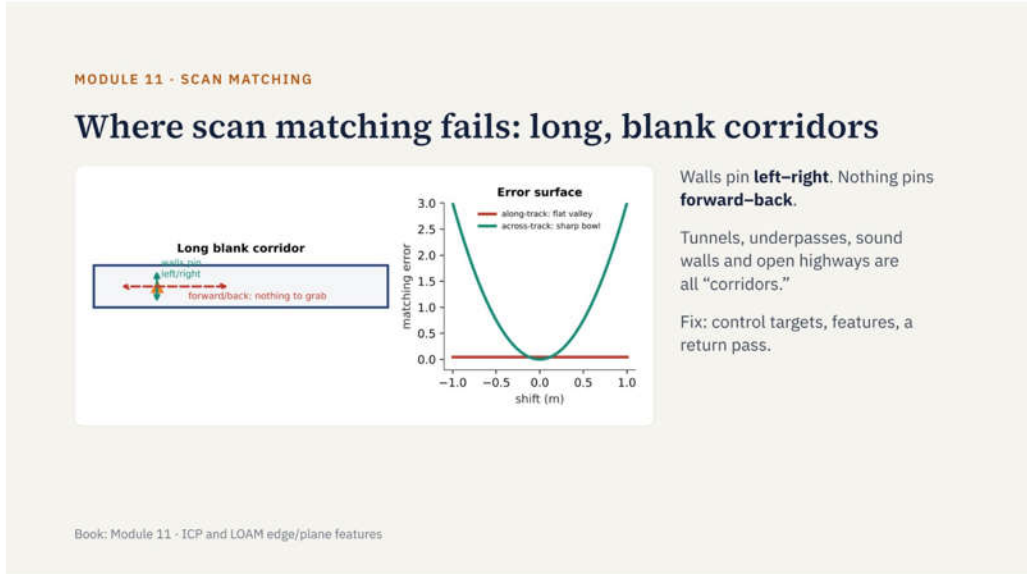
Edges (corners, poles, curb lips) and **planes** (walls, pavement) are kept; the rest is skipped.

Fewer points, faster matching — LIO-SAM processes a scan in 42–101 ms.

No edges or planes in view = nothing to match.

Book: Module 11, Examples 11.2–11.3 · LOAM default: c > 0.1 edge, c < 0.1 plane

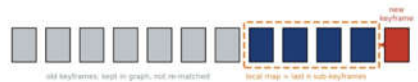
Flat wall: neighbor ranges 10.00, 10.01, 10.00, 9.99, 10.00 — differences cancel, c = 0, a plane. Corner: 10, 10, 8, 8, 8 — c = 0.125, an edge. The matcher then minimizes the distance of each edge point to the nearest edge line and each plane point to the nearest plane patch. This is why scene content — not scanner price — often decides SLAM quality.



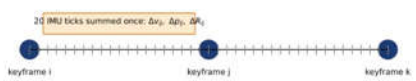
Scan matching (ICP, LOAM) slides a new scan until it fits the map. On the right of the figure: across-track the fit has a sharp minimum, along-track it is a flat valley — any shift fits equally well. That is why underpasses are the weak link on every corridor project and why the capstone puts a control target in the middle of one.

MODULE 12 · LIO-SAM BUILDING BLOCKS

Keyframes and IMU preintegration



A new **keyframe** is matched to a local map built from the last few — not to every scan ever taken.



All IMU samples between keyframes are **summed once** into one change of position, velocity and rotation.

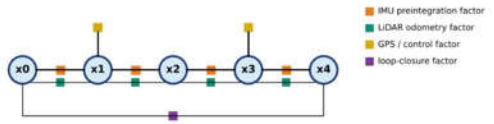
Book example: 10 samples at 0.50 m/s² over 0.1 s → Δv = 0.050 m/s, Δp = 2.5 mm. Ten numbers became two.

Book: Modules 7 and 12, Example 7.4 · Source: Shan et al., LIO-SAM (2020)

These two ideas make real-time SLAM possible. Keyframes keep the graph small; preintegration turns hundreds of IMU readings into one measurement between keyframes, which can be re-used when the bias estimate changes. The optimizer then treats it like any other observation with a weight.

MODULE 12 · FACTOR GRAPHS

A factor graph is a least-squares adjustment



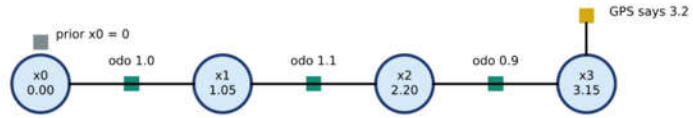
Factor	What it measures	Survey analog
IMU preintegration	motion between poses	dead reckoning
LiDAR odometry	scan-to-map fit	traverse leg
GNSS	absolute position	control point
Loop closure	return to a known place	closing the loop

Book: Module 12, Example 12.1 — a 0.20 m GPS misclosure shared equally by four measurements

This is the slide that makes SLAM familiar to surveyors. Each pose is an unknown; each sensor contributes an observation equation with a weight; the optimizer solves the normal equations. The 1-D example in the book distributes a 0.20 m misclosure across three odometry legs and a GPS fix exactly like a least-squares traverse adjustment.

MODULE 12 · A POSE GRAPH BY HAND

The misclosure is shared

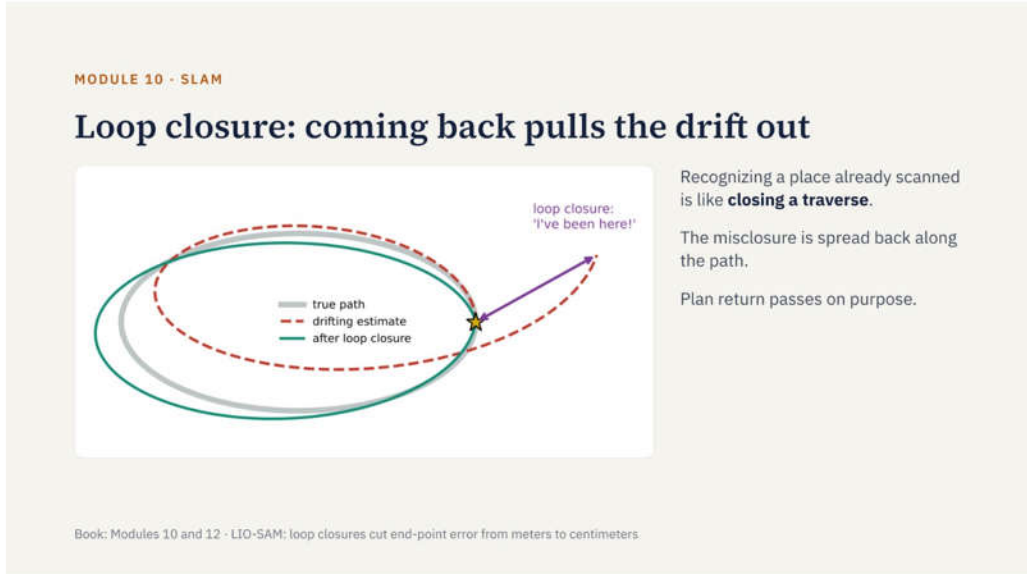


odometry adds to 3.0, GPS says 3.2 -> share the 0.2 m disagreement

Measurement	Observed	Adjusted	Residual
Odometry legs	1.0 + 1.1 + 0.9 = 3.0 m	1.05, 1.15, 0.95	+0.05 each
GPS at x3	3.2 m	3.15 m	−0.05

Book: Module 12, Example 12.1 · Workbook sheet M12

Equal weights, so the 0.20 m disagreement is split into four equal residuals of 0.05 m — exactly a least-squares traverse adjustment. In a real system the weights come from each sensor’s uncertainty: a weak scan match gets less say, an RTK fix more. Ask the room: what would change if GPS had σ = 0.02 m instead of 0.10 m? Answer: the poses move closer to 3.2 and the odometry residuals grow.



SLAM — simultaneous localization and mapping — builds the map and finds the vehicle in it at the same time. Drift accumulates; recognizing a previous place lets the optimizer distribute the misclosure. The LIO-SAM paper shows end-to-end errors falling from meters to 4–17 cm when loop closures and GPS are added. Caution: a closed loop can still be rotated or scaled — we will see that in the GX5 traverse.

MODULE 12 · LIO-SAM RESULTS

Each factor removes a layer of drift

Dataset	LOAM	LIOM	LIO-odom	LIO-GPS	LIO-SAM
Campus, 1.4 km (0.9 mi)	192.43 m	fail	9.44 m	6.87 m	0.12 m
Park, 2.9 km (1.8 mi)	121.74 m	34.60 m	36.36 m	2.93 m	0.04 m
Amsterdam, 19 km (11.8 mi)	fail	fail	fail	1.21 m	0.17 m

LIO-odom: IMU + LiDAR only

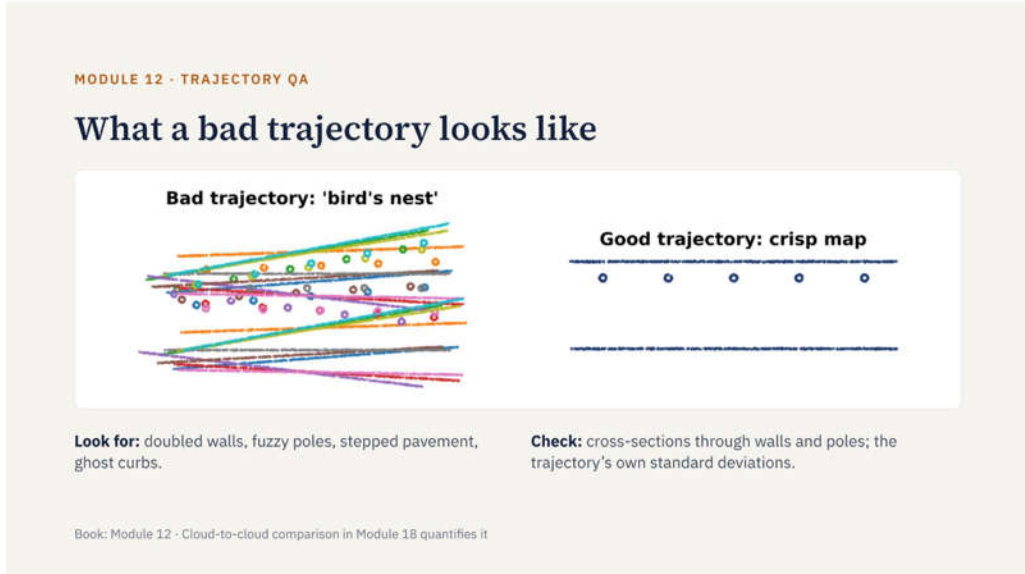
LIO-GPS: + GPS factors

LIO-SAM: + loop closures

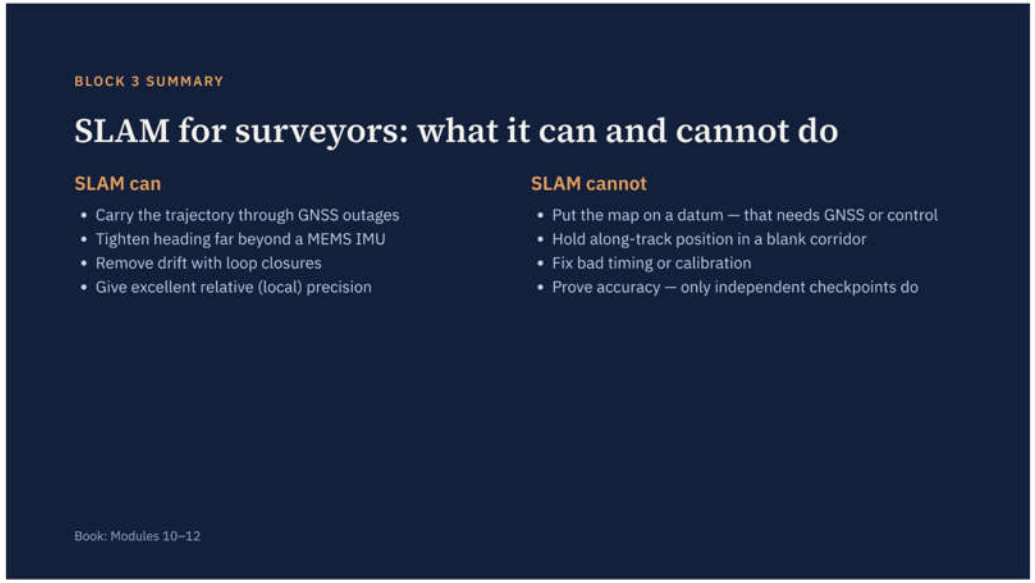
End-to-end translation error. Research results on a robot — **not survey-grade checkpoint accuracy**.

Source: Shan et al., "LIO-SAM: Tightly-coupled Lidar Inertial Odometry via Smoothing and Mapping," IROS 2020, Table II

Read across a row: LiDAR alone (LOAM) drifts tens to hundreds of meters; adding the IMU helps; adding GPS helps more; adding loop closures brings the error to 4–17 cm over kilometers. That is excellent odometry, but it was judged against start/end points or GPS, not independent checkpoints. A deliverable still needs calibration, control and an ASPRS test. Note also: the paper's eq. (12) writes the relative transform with a transpose; the correct general form uses the inverse.



A quick visual QA any surveyor can do: slice the cloud through a building face or a pole. One crisp line means the passes agree; two or more lines — the bird’s nest — mean the trajectory disagrees between passes. Find where it starts and look at the trajectory report there: usually an outage, a corridor, or a false loop closure.



Close the block with this slide and take questions. The planning rules that follow: map where GNSS is blocked and where scenes are featureless; put control and checkpoints there; drive return passes for loop closures; read the trajectory report before the point cloud; and never let a SLAM residual stand in for an independent accuracy test.

2:40–3:10

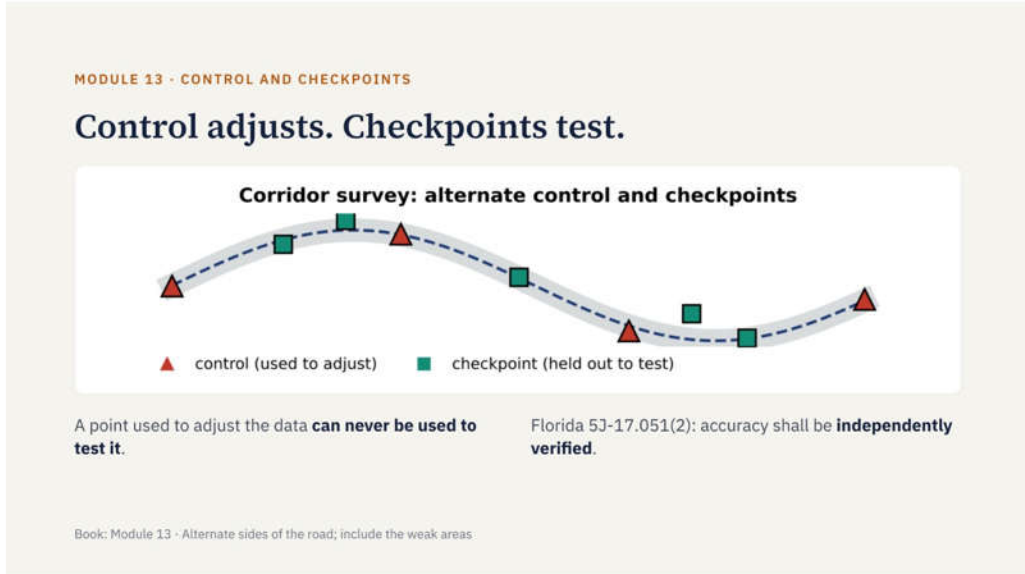
Lunch

Over lunch: find the stretch of your next project where GNSS is blocked and the scene is blank. That is where SLAM needs control.

Thirty minutes. After lunch: how to test and report the accuracy SLAM produced.



45 minutes. Now that we know how the trajectory is built, two questions: how do I prove the accuracy of a dataset (ASPRS testing), and how do I predict it before I collect (error budgets)?



This is the single most important rule in accuracy testing. Control enters the adjustment; checkpoints are held out. Using a control point as a checkpoint tests nothing — it just confirms the adjustment did what you told it to. Distribute checkpoints along the whole project, both sides, and deliberately in the weak areas like the underpass.



Walk through the five rules. Rule 4 is new in Edition 2 and surprises people: checkpoint survey error is added into the reported RMSE, so a 1.9 cm data fit with 0.8 cm checkpoints reports as 2.06 cm. NVA = non-vegetated vertical accuracy; VVA = vegetated vertical accuracy. Ten points is fine for a field check, but say so in the statement.

MODULE 13 - DATA INTERNAL PRECISION

Passing the checkpoints is not enough

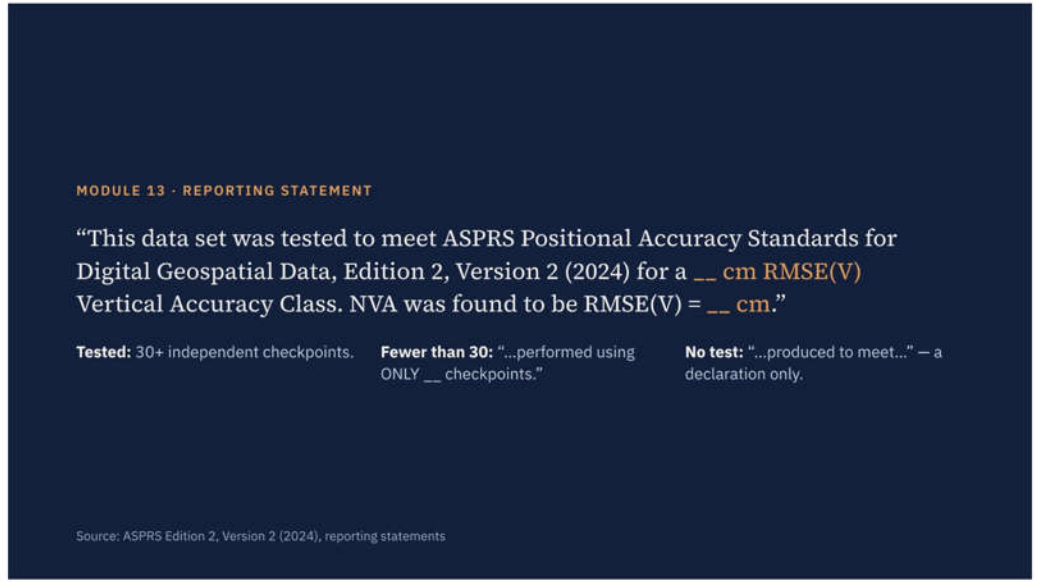
Vertical class	NVA RMSE	Within-swath max diff.	Swath-to-swath RMSD	Swath-to-swath max
Rule	≤ X	≤ 0.60X	≤ 0.80X	≤ 1.60X
5 cm (0.16 ft)	5 cm / 0.16 ft	3 cm / 0.10 ft	4 cm / 0.13 ft	8 cm / 0.26 ft
10 cm (0.33 ft)	10 cm / 0.33 ft	6 cm / 0.20 ft	8 cm / 0.26 ft	16 cm / 0.52 ft

Within-swath: is a flat parking lot flat in one pass? Tests range noise.

Swath-to-swath: do overlapping passes agree? Tests trajectory and calibration.

Source: ASPRS Edition 2, Version 2 (2024), Table 7.2

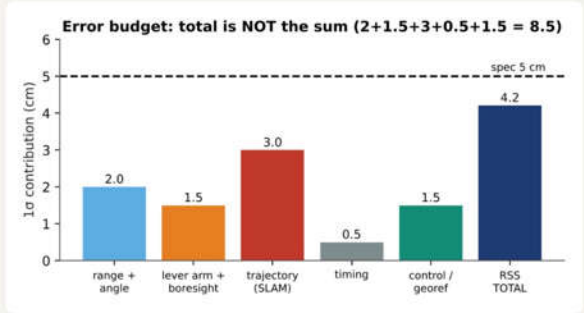
A project can pass NVA at the checkpoints and still have passes that disagree between them. That is why Edition 2 adds internal precision checks for LiDAR. The swath-to-swath check is the cloud-to-cloud comparison we will see in Block 6.



The words matter. “Tested to meet” requires the independent test. “Produced to meet” is only a declaration by the producer — and calibration control cannot be reused as checkpoints. In Florida, the statement goes on the Survey and Map Report along with the datums.

MODULE 14 · ERROR BUDGETS

The total is not the sum



Independent errors combine by **root-sum-square**: 4.2 cm (0.14 ft), not 8.5 cm.

The biggest term dominates. Here: **trajectory**.

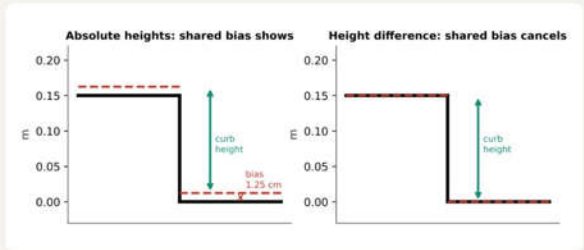
Trajectory at 5 cm → total 5.8 cm: **fails**.

Book: Module 14, Example 14.8 · Workbook sheet M14

Square each term, add, take the square root. Because the largest term dominates, the way to fix a failing budget is to attack the biggest term — here the trajectory: more control in weak stretches, slower speed, a return pass for loop closures. Buying a better laser barely moves the total. Demonstrate live in the workbook: sheet M14, Example 14.8, change trajectory from 3.0 to 5.0.

MODULE 14 · CORRELATION

Relative measurements beat absolute ones



Each point: ±1.3 cm absolute.

Curb height from the same sweep: **±0.35 cm** — the shared bias cancels.

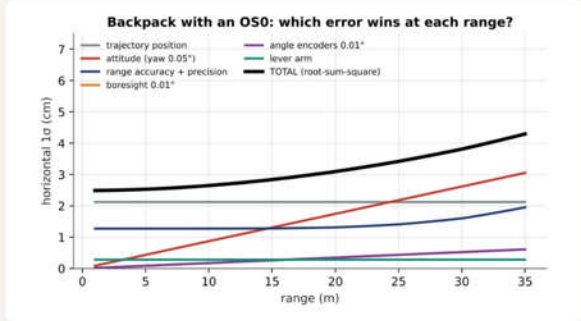
Cross-slopes, curb reveals and cracks map better than the datasheet suggests.

Book: Module 14, Example 14.2

The OS0 has a 1.25 cm range bias plus 0.25 cm noise at short range. Both curb top and gutter share the same bias, so the difference — the curb height — keeps only the noise. The reverse is also true: for volumes, a vertical bias multiplies by the area — 1.25 cm over 2,000 m² is 25 m³ — so for stockpiles, bias is everything.

MODULE 14 · BUDGET BY RANGE

Which error wins depends on range

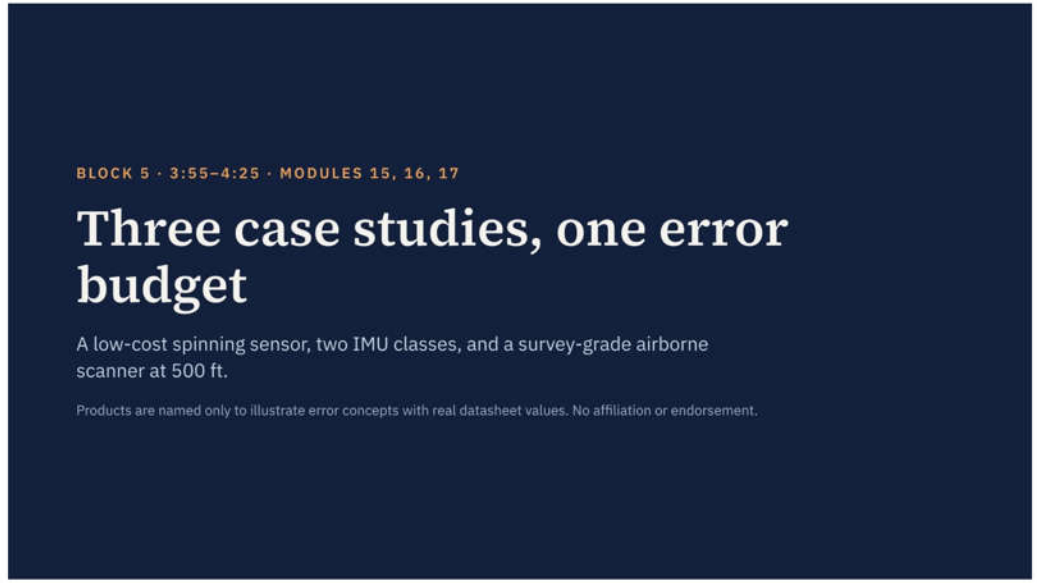


Range	H 1σ	V 1σ
5 m (16 ft)	2.5 cm	2.0 cm
10 m (33 ft)	2.7 cm	2.1 cm
20 m (66 ft)	3.1 cm	2.2 cm
35 m (115 ft)	4.3 cm	2.5 cm

Close in: trajectory. Far out: **heading**.

Book: Module 14, Examples 14.4 and 14.9 · Workbook sheet M14

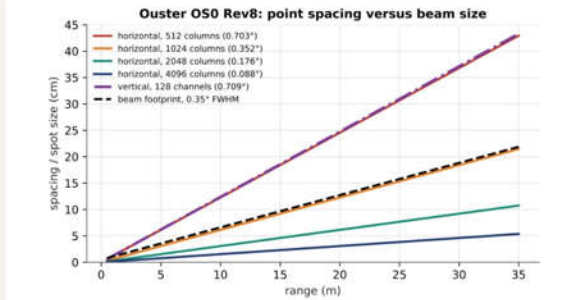
Backpack OS0 with good SLAM and control: trajectory 1.5/1.5/2.0 cm, yaw 0.05°, roll/pitch 0.02°. The trajectory term is flat; the yaw term grows linearly with range and overtakes near 25 m. Practical rule: map hard features within 15–20 m for 2–3 cm work. The laser itself is only the third-largest term.



30 minutes. Read the neutrality statement aloud — it matters for CE approval. Each case study applies the Module 14 budget to a real datasheet. The lesson is the same in all three: find the dominant term.

MODULE 15 · CASE STUDY: OUSTER OS0 REV8

More points are not sharper points



At 20 m (66 ft) the spot is **12.7 cm**.

4096 mode spaces points 3 cm apart — four samples inside every spot.

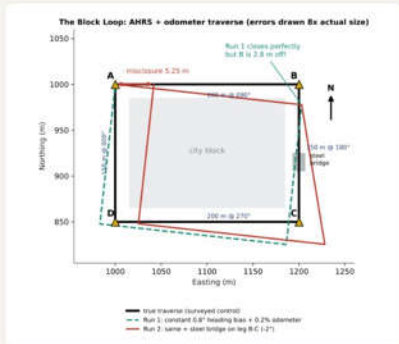
Range bias 1.25 cm is a floor only calibration removes.

Source: Ouster OS0 Rev8 datasheet · Book: Module 15

Lessons from the OS0: (1) the 0.35° beam footprint, not the column count, limits edge sharpness; (2) on a 10% target precision degrades past about 25 m; (3) never use the Low Data Rate profile for survey work — its 0.8 cm quantization adds 36% noise; (4) synchronize with PTP or PPS — under 1 ms is fine walking, marginal driving.

MODULE 16 · CASE STUDY: MICROSTRAIN 3DM-GX5

A perfect closure can still be wrong



Run 1: 0.8° heading bias on every leg. Misclosure: zero. Point B: **2.8 m (9.3 ft) off**.

Run 2: a steel bridge deflects the magnetometer 2°. Misclosure 5.25 m — 1:133.

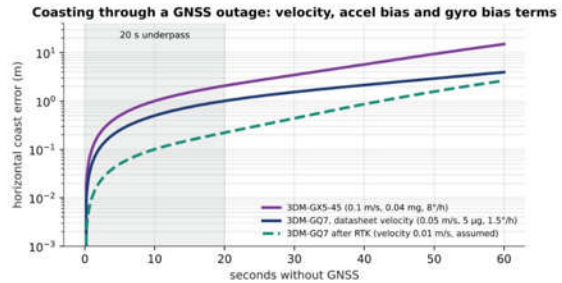
A rotated, stretched loop still closes. Only a **second control point** reveals it.

Source: MicroStrain 3DM-GX5-25/-45 datasheets · Book: Module 16 · Workbook sheet M16

This is the most memorable slide of the case studies for surveyors, because it is a traverse. An AHRS with a constant heading bias produces a figure that closes perfectly — rotated and scaled — so the closure check gives false comfort. The fix is the same as in conventional surveying: check into a second known point or an azimuth mark. Compass-rule adjustment of Run 2 makes it look closed, but B and C remain about 3 m wrong.

MODULE 16 · GX5-45 VERSUS GQ7 WITH RTK

How far does each IMU coast?



System	20 s	60 s
GX5-45	2.1 m	15 m
GQ7	1.0 m	—
GQ7 + RTK	0.22 m	2.6 m

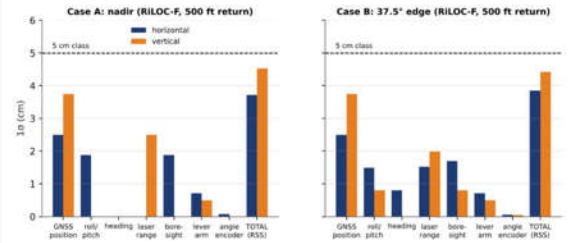
No MEMS INS holds centimeters through a minute-long tunnel. That takes SLAM, control, or a navigation-grade INS.

Source: MicroStrain 3DM-GX5-45 and 3DM-GQ7 datasheets · GQ7 RTK via FPRN (free, NTRIP)

The better gyro (1.5°/h versus 8°/h) wins on long outages because the gyro term grows with time cubed. RTK-quality velocity wins on short outages. In Florida, choose an FPRN network (VRS or iMAX) mountpoint rather than the nearest single station: at 22 km the 1 ppm term alone adds 2.2 cm. Dual-antenna heading needs a long baseline — on a backpack it is no better than a magnetometer.

MODULE 17 · CASE STUDY: RIEGL VUX-240²⁴ AT 500 FT

Survey grade: GNSS is the biggest term

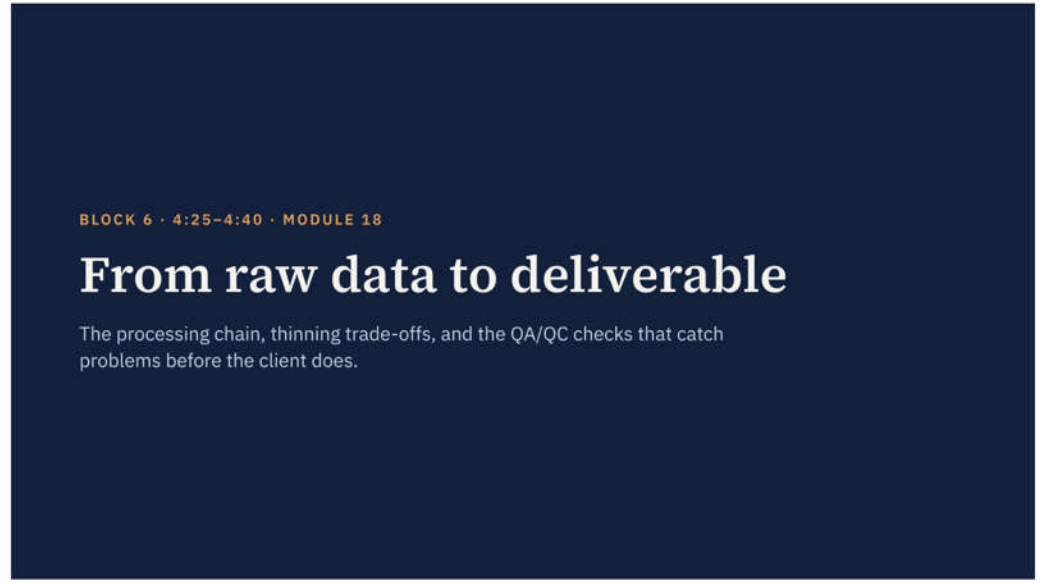


RiLOC-F	H	V
Direct, nadir	3.7 cm	4.5 cm
Direct, edge	3.9 cm	4.4 cm
With control	3.1 cm	3.1 cm

Laser: 20 mm accuracy / 15 mm precision at 150 m. Attitude 0.005° ≈ 1.3 cm at 500 ft.

Source: RIEGL VUX-240²⁴ datasheet (2026-08-17) · Book: Modules 14 and 17 · 1σ values

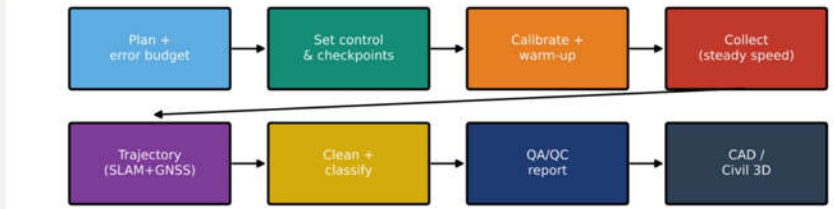
At 500 ft (152.4 m) the RiLOC-F attitude of 0.005° projects to 1.3 cm. Direct georeferencing alone gives about 3.7 cm horizontal and 4.5 cm vertical: under 5 cm, but with little vertical margin. With ground control and strip adjustment both drop to about 3 cm. Also: at 2.4 MHz the unambiguous range is only 62 m, so returns from 500 ft are in MTA zone 3 — ghost layers are blunders, not noise. Atmospheric correction at this range is 4.6 cm — confirm it is on.



15 minutes. Keep it practical: what each step does, where accuracy can be lost, and which checks go in the report.

MODULE 18 · PROCESSING

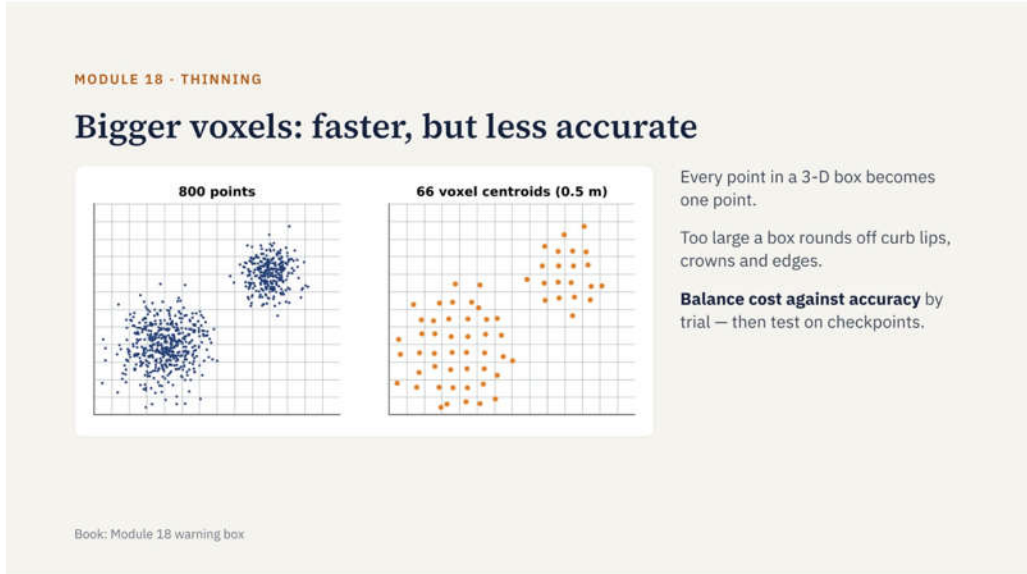
The project chain, plan to CAD



Log every setting — filters, voxel size, classification parameters, software versions. They are part of the **measurement records** (5J-17.050(8)).

Book: Module 18 · LAS 1.4 / LAZ is the usual delivery format

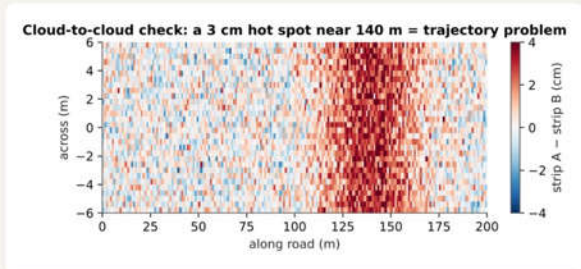
Walk the chain: plan with an error budget; set control and checkpoints; calibrate and warm up the IMU; collect at a steady speed; process the trajectory (SLAM plus GNSS, base or CORS data); clean (statistical outlier removal) and classify to ASPRS classes; QA/QC report; then CAD or Civil 3D. The processing log is what lets you — or a reviewer — reproduce the product, which Florida's records rule requires.



Voxel thinning is the most common hidden accuracy loss in processing. A 10 cm voxel is fine for a pavement surface model and wrong for a curb line. There is no universal setting: try two or three sizes on a test area, compare against checkpoints and against the full-density cloud, and record the choice.

MODULE 18 · QA/QC

Cloud-to-cloud: do the passes agree?



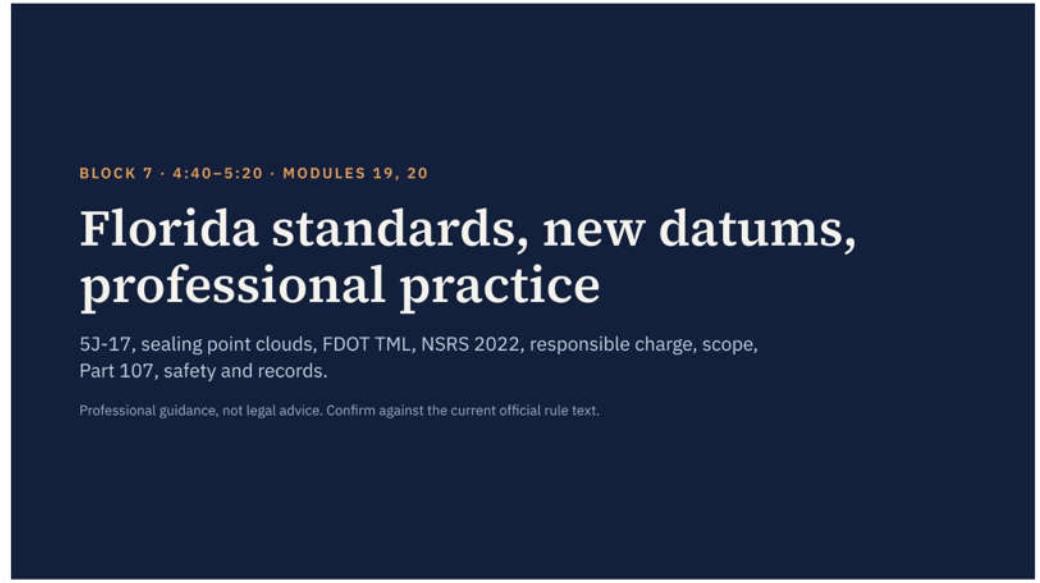
Compare northbound against southbound on hard surfaces.

A band like this 3 cm hot spot near 140 m is a trajectory problem — found between the checkpoints.

This is ASPRS swath-to-swath precision.

Book: Module 18 · Capstone result: mean 0.2 cm, spread 0.9 cm, one patch at the underpass exit

Checkpoints test a few dozen locations; cloud-to-cloud tests everywhere the passes overlap. In the capstone, one 2.5 cm patch at the underpass exit was flagged and re-checked against an extra shot. Report what you found and what you did about it.



40 minutes. This is where the seminar earns its standards-related CE content. Rule citations are paraphrased; have the current Chapter 5J-17 text available. Before each offering, re-check the NGS modernization timeline and the FDOT Surveying and Mapping Handbook edition.

MODULE 19 · RULE 5J-17.050 DEFINITIONS

LiDAR is inside the Standards of Practice

Rule	Defines	For a mobile LiDAR survey
.050(10)	Remote sensing	Mobile LiDAR is remote sensing
.050(11)	Survey	Includes direct or remote sensing methods
.050(7)	Map of survey	Graphical or digital: a delivered point cloud or surface is a map
.050(8)	Measurement records	Raw sensor data, trajectory, calibration
.050(12)	Survey and Map Report	Identifies the responsible surveyor and mapper
.050(13)	Work product	Made by or under the PSM in responsible charge

Source: Fla. Admin. Code R. 5J-17.050 (paraphrased) · Book: Module 19

The key point: there is no exemption for remote sensing. A point cloud delivered as a survey product is a map of survey, the raw data and trajectory are measurement records, and the whole thing is a work product under a PSM in responsible charge.

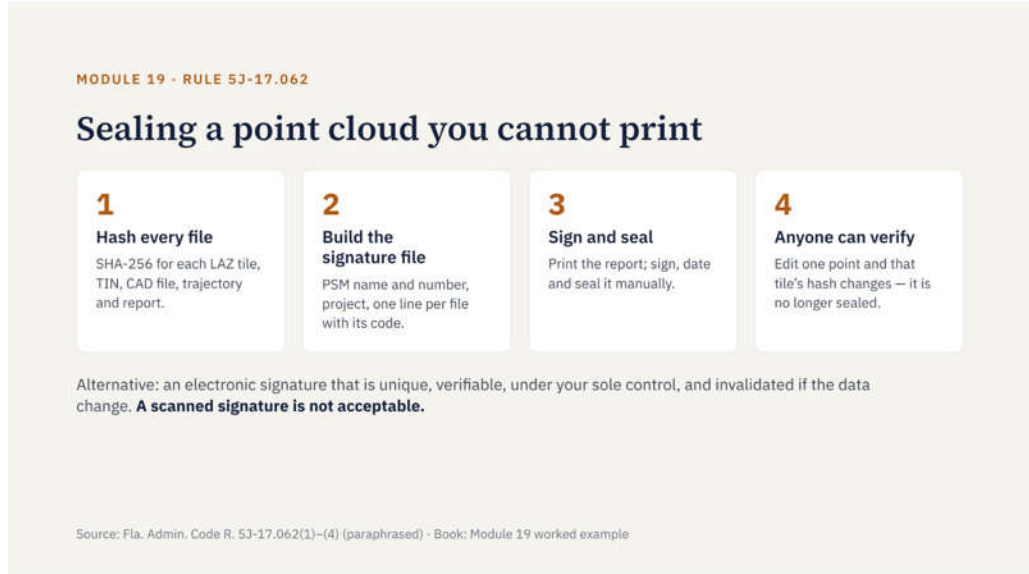
MODULE 19 · RULE 5J-17.051 GENERAL REQUIREMENTS

What the deliverable must carry

Rule	Requirement, applied to mobile LiDAR
.051(2)	Accuracy independently verified — independent checkpoints, RMSE in ASPRS Edition 2 terms
.051(2)(a)	Vertical datum and control stated, e.g. NAVD 88 via GEOID18; NVA test documented
.051(2)(b)	Horizontal datum stated, e.g. NAD 83(2011) epoch 2010.0, SPCS Florida East
.051(2)(d)	Records dated to acquisition: raw data, trajectory, calibration, processing logs
.051(3)(d)	Survey date = last date of data acquisition, not the processing date
.051(3)(i)	Merged data from another firm: state who is primarily responsible
.052	Boundary requirements still apply — LiDAR topo is not a boundary survey

Source: Fla. Admin. Code R. 5J-17.051 and .052 (paraphrased) · Book: Module 19

Highlight three items people miss: (1) the survey date is the last acquisition date, not the date the cloud was processed; (2) when you merge another firm's LiDAR into your survey, say who is responsible for what; (3) 5J-17.052 still governs boundaries — a LiDAR topographic survey does not satisfy boundary requirements.



A multi-gigabyte point cloud can be sealed without printing it: the signature file of secure authentication codes is what gets sealed. Practical consequence: any later edit — even reclassifying a few points — means a new hash and a new sealed signature file. Tools: sha256sum on Linux or Mac, certutil -hashfile on Windows.

MODULE 19 · FDOT TERRESTRIAL MOBILE LIDAR (TML)

On FDOT work, the Handbook governs

Type	Typical use	Accuracy (from App. C)
A	Design engineering, as-built	_____
B	Asset inventory, earthwork	_____
C	Planning, statistics	_____

Fill in from the Handbook edition named in your contract.

FDOT Surveying and Mapping Handbook (Feb 28, 2025), TML Sections 35–40 and Appendix C.

§39.6 control and checkpoints · §39.7 accuracy analysis · §39.8 QA/QC · §40 deliverables and the TML survey report.

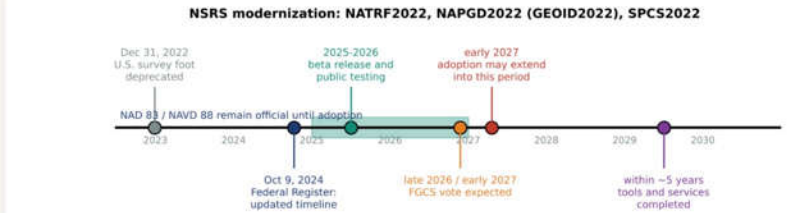
Check your District's QA/QC plan (e.g. D5: MSTS number, Central Office submittal).

Source: FDOT Surveying and Mapping Handbook (Feb 28, 2025); FDOT District 5 QA/QC plan · Book: Module 19

Instructor: before presenting, fill the three accuracy cells from Section 39.6.2 and Appendix C of the current Handbook — the seminar deliberately does not print numbers that could go out of date. Types A, B, C originate in FDOT's 2012 TML guidelines adapted from Caltrans. In District 5, imagery including TML needs a Mobile Survey Tracking System number and the raw, interim and final data go to the Central Office.

MODULE 19 · NSRS MODERNIZATION

New datums are coming — not here yet



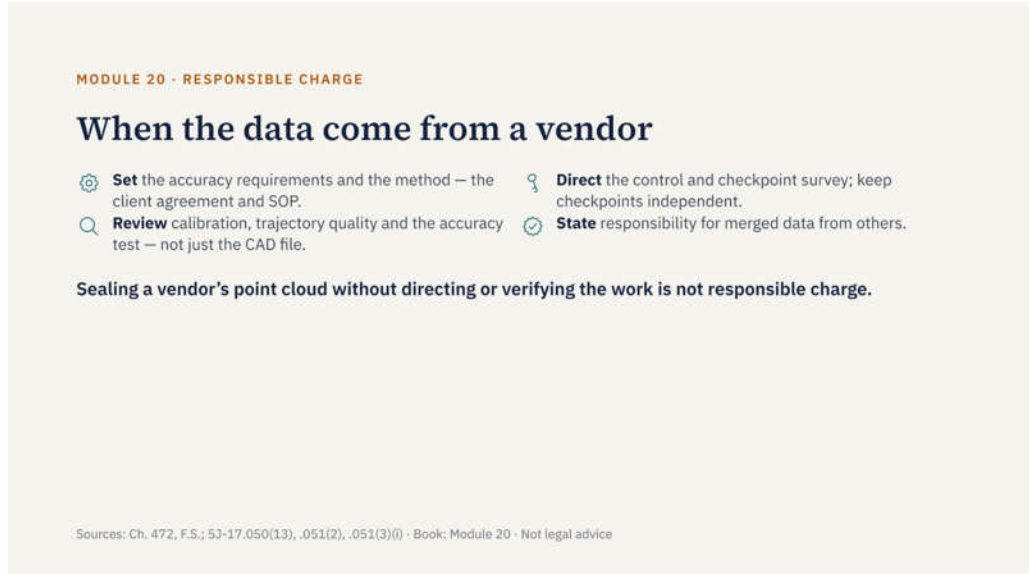
NATRF2022	NAPGD2022 / GEOID2022	SPCS2022
replaces NAD 83	replaces NAVD 88 and GEOID18	meters; international foot only

Sources: Federal Register, Oct 9, 2024; NGS NSRS timeline · Until adoption, NAD 83 and NAVD 88 remain official · Re-check before each offering

Instructor: check the NGS “NSRS Modernization Timeline” page the week of the seminar — dates after 2026 are NGS expectations, not fixed. Contract advice: state who pays for later transformation of deliverables to NATRF2022, and keep the raw data and trajectory so it can be re-processed.



U.S. survey foot = 1200/3937 m = 0.3048006096 m; international foot = 0.3048 m. On 650,000 ft the difference is 0.396 m, or 1.30 ft. The U.S. survey foot was deprecated at the end of 2022, and SPCS2022 will use the international foot only. Florida law (s. 177.151 F.S.) speaks of feet and decimals of a foot — so label which foot, every time.



Ask the room: a vendor delivers a classified point cloud and you never saw the checkpoints. What must you do before sealing it? Answer: direct or perform an independent checkpoint survey, review calibration and trajectory reports, run the accuracy test yourself, and state responsibility for the vendor's data — or deliver it unsealed, clearly labeled for information only.

MODULE 20 · CONTRACT SCOPE AND LIABILITY

Write it down before you drive

Item	What the contract or agreement should say
Accuracy	H and V classes (RMSE, ASPRS Edition 2), surfaces tested, checkpoint count
Limitations	No view through solids, dense vegetation, water (NIR) or underground; occlusions
Intended use	e.g. design topography, not boundary; no reuse without review
Datums and units	Datum, realization, epoch, geoid, which foot; who pays to move to NATRF2022
Third-party data	Identify it, its accuracy, and your reliance on it
Data license	Ownership of raw and processed data; reuse, resale, retention
Changes	Added scope, recollection after construction, new accuracy demands

Book: Module 20 and Appendix E (sample SOP and client accuracy agreement) · Not legal advice

Appendix E has a sample client accuracy agreement with horizontal and vertical classes the client signs. Most disputes on LiDAR projects come from unstated limitations (we did not see under the water, behind the parked truck, under the canopy) or unstated intended use. Also confirm professional liability coverage fits the contract.



Flying the RIEGL at 400 ft instead of 500 ft is both legal without an altitude waiver and slightly more accurate — the attitude terms scale with range. Waivers take time (the FAA generally responds within about 90 days), so plan early. On roads, a maintenance-of-traffic plan is part of the job; never set targets in live lanes without protection. Airborne scanners like the VUX-240[24] are laser Class 3R with a nominal ocular hazard distance; follow the manufacturer’s limits.

MODULE 20 · RECORDS AND RETENTION

Keep what lets you rebuild the survey

Record	Why keep it
Raw LiDAR, IMU, GNSS, camera	Reconstructs the measurements (5J-17.050(8)); OS0 ≈ 133 GB/hour
Base station / CORS logs	Reprocess the trajectory; prove the datum
Trajectory with standard deviations	Shows where accuracy is strong or weak
Calibration file and drive	Ties lever arm and boresight to the data
Control and checkpoint survey	Proves the accuracy statement
Processing logs and settings	Reproducibility — voxel size, filters, versions
Deliverables and sealed signature file	What the client received

Two copies in two places. Use the sealing hashes to prove archives are unchanged.

Book: Module 20 · Retention period: contract, client policy and law (public-records schedules for agencies)

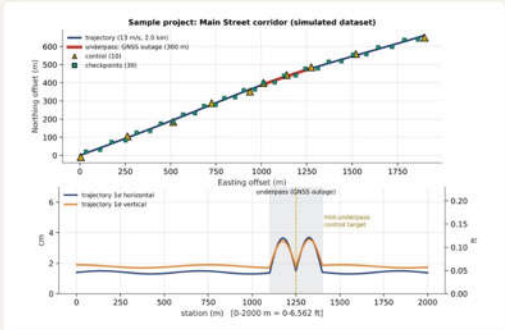
Raw data volumes are large — plan storage in the proposal. The checksums you computed for sealing double as archive integrity checks. Keeping raw data and trajectories is also what makes a future transformation to the new datums possible without re-collecting.



20 minutes. Walk through the project as a reviewer would: trajectory first, then checkpoints, then the accuracy report. Everything here ties back to the earlier blocks — speed from point spacing, the underpass coast, the vertical error budget, and the ASPRS test.

MODULE 21 · STEP 1: READ THE TRAJECTORY

Read the trajectory before the cloud



Open sky: **1.4 cm (0.05 ft)** horizontal 1σ.

Underpass: rises to **3.6 cm (0.12 ft)**, pulled down by the mid-underpass control target.

Put extra checkpoints where the trajectory is weak.

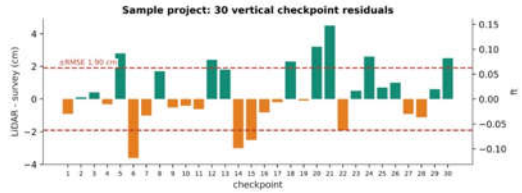
Speed: 13 m/s (≈ 30 mph) gives 0.16 m line spacing — under the 0.25 m requirement.

Book: Module 21 · Data: sample_trajectory.csv · Workbook: Dataset sheet

Plan recap: 300 m underpass at 13 m/s is 23 seconds without GNSS; an unaided tactical IMU would drift 1.3 m along-track — so the plan adds chevron targets at entrance, midpoint and exit plus a return pass. The trajectory report confirms the plan worked: the bump in the underpass is held to 3.6 cm. Vertical budget: 2.45 cm open sky, 3.00 cm in the underpass without the extra target, 2.23 cm with it.

MODULE 21 · STEP 2: TEST THE CHECKPOINTS

30 checkpoints, no blunders



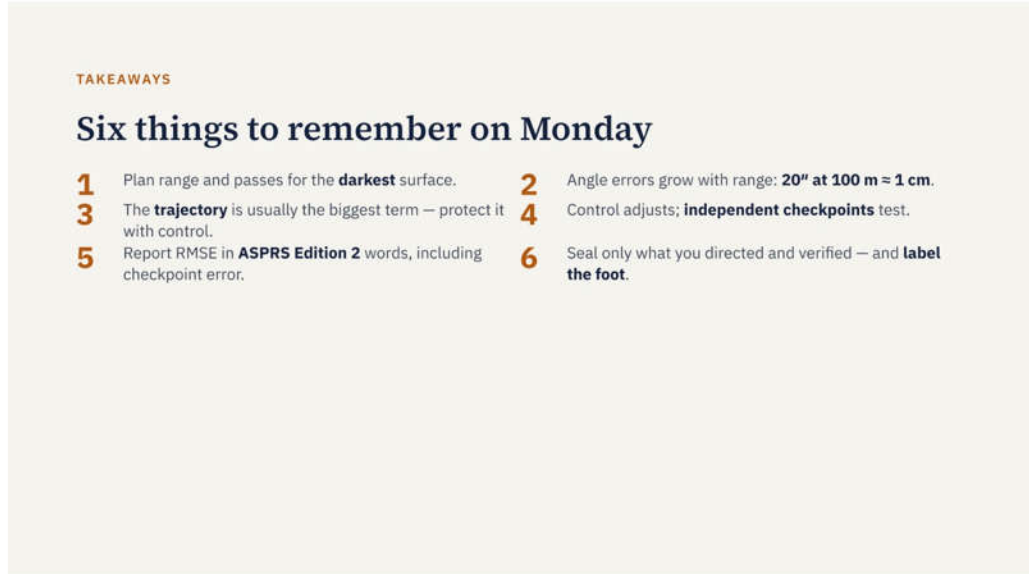
Mean dZ	RMSE z (data)	RMSE H (data)	Largest |dZ|	3 × RMSE
+0.31 cm	1.90 cm	2.63 cm	4.5 cm	5.7 cm
+0.010 ft	0.062 ft	0.086 ft	0.148 ft	0.187 ft

Book: Module 21 checkpoint table · Data: sample_checkpoints.csv · Workbook: Dataset sheet

Read the plot like a reviewer: is there a bias (mean +0.31 cm — negligible)? Any blunders (largest residual 4.5 cm, under three times RMSE, 5.7 cm — none flagged)? Any pattern along the corridor, such as a cluster at the underpass? Checkpoints alternate sides of the road and include the underpass.



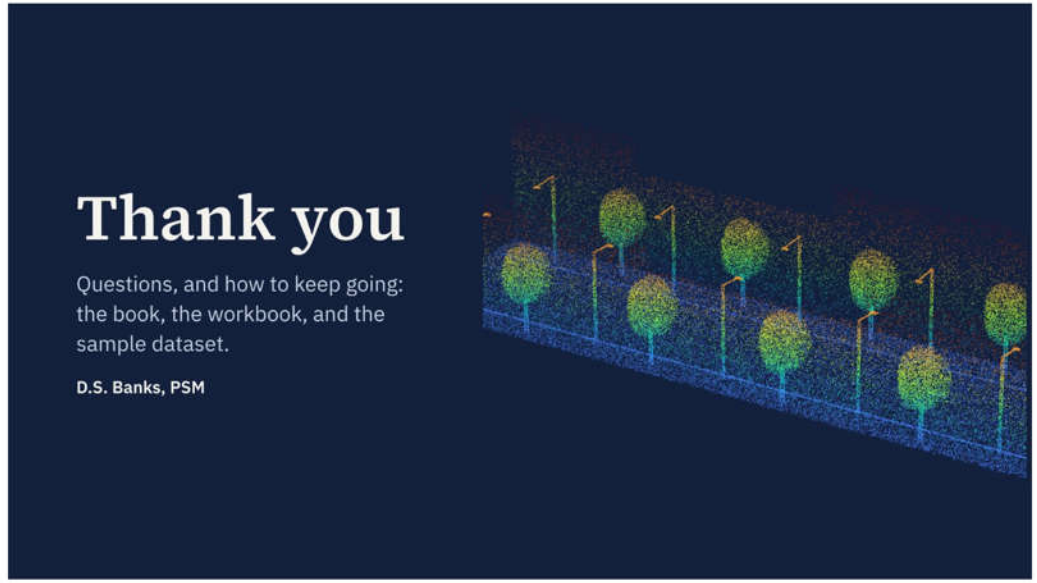
Final numbers: $\sqrt{(1.90^2 + 0.8^2)} = 2.06$ cm vertical; $\sqrt{(2.63^2 + 1.0^2)} = 2.82$ cm horizontal. The Dataset sheet in the workbook builds this statement automatically from the 30 checkpoints. In Florida, the Survey and Map Report also carries the datums (NAD 83(2011) epoch 2010.0, NAVD 88 via GEOID18), the foot definition, the last acquisition date, limitations, and the PSM's seal.



Two minutes. Ask each participant to pick the one takeaway most relevant to their next project. Then hand out the exam.



Instructor: distribute Version A and Version B alternately by seat. Keys and objective maps are in the instructor packet. Collect evaluations with the exams.



Open the floor for questions. Remind participants that the workbook lets them rerun every example with their own equipment specs.